
EECS 16A Designing Information Devices and Systems I

Discussion 4A

1. Mechanical Inverses

A square matrix A is said to be invertible if there exists a matrix B such that $AB = I$ and $BA = I$. The matrix B is called the inverse of A and is denoted by A^{-1} .

Some properties of the inverse are:

- $(A^{-1})^{-1} = A$
- $(A^T)^{-1} = (A^{-1})^T$

In each part, determine whether the inverse of $\mathbf{A}$ exists. If it exists, find it.

(a) $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}$

(b) $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

(c) (PRACTICE)

$$\mathbf{A} = \begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix}$$

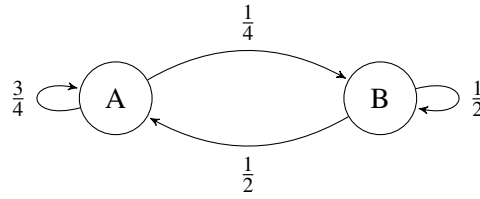
(d) $\mathbf{A} = \begin{bmatrix} 5 & 5 & 15 \\ 2 & 2 & 4 \\ 1 & 1 & 4 \end{bmatrix}$

(e) (PRACTICE)

$$\mathbf{A} = \begin{bmatrix} 5 & 5 & 15 \\ 2 & 2 & 4 \\ 1 & 0 & 4 \end{bmatrix}$$

2. Transition Matrix

Suppose we have a network of pumps as shown in the diagram below. Let us describe the state of A and B using a state vector $\vec{x}[n] = \begin{bmatrix} x_A[n] \\ x_B[n] \end{bmatrix}$ where $x_A[n]$ and $x_B[n]$ are the states at time-step n .



- Find the state transition matrix S , such that $\vec{x}[n+1] = S \vec{x}[n]$.
Separately find the sum of the terms for each column vector in S . Do you notice any pattern?
- Let us now find the matrix S^{-1} such that we can recover the previous state $\vec{x}[n-1]$ from $\vec{x}[n]$.
Specifically, solve for S^{-1} such that $\vec{x}[n-1] = S^{-1} \vec{x}[n]$.
- Now draw the state transition diagram that corresponds to the S^{-1} that you just found.
Also find the sum of the terms for each column vector in S^{-1} . Do you notice any pattern?
- Redraw the diagram from the first part of the problem, but now with the directions of the arrows reversed. Let us call the state transmission matrix of this "reversed" state transition diagram T . Does $T = S^{-1}$?
- Suppose we start in the state $\vec{x}[1] = \begin{bmatrix} 12 \\ 12 \end{bmatrix}$. Compute the state vector after 2 time-steps $\vec{x}[3]$.
- (Challenge practice problem)** Given our starting state from the previous problem, what happens if we look at the state of the network after a lot of time steps? Specifically which state are we approaching, as defined below?

$$\vec{x}_{final} = \lim_{n \rightarrow \infty} \vec{x}[n]$$

Note that the final state needs to be what we call a *steady state*, meaning $S \vec{x}_{final} = \vec{x}_{final}$.

Also what can you say about $x_A[n] + x_B[n]$?

Use information from both of these properties to write out a new system of equations and solve for $\vec{x}_{final}$.